The Tinkerers

The Tinkerers | 2026

Server-Side WebAssembly in Practice

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1. Agenda

01
WebAssembly & WASI Basics

02
Developer experience: runtimes, tooling, components

03
HTTP servers & AI with WASM

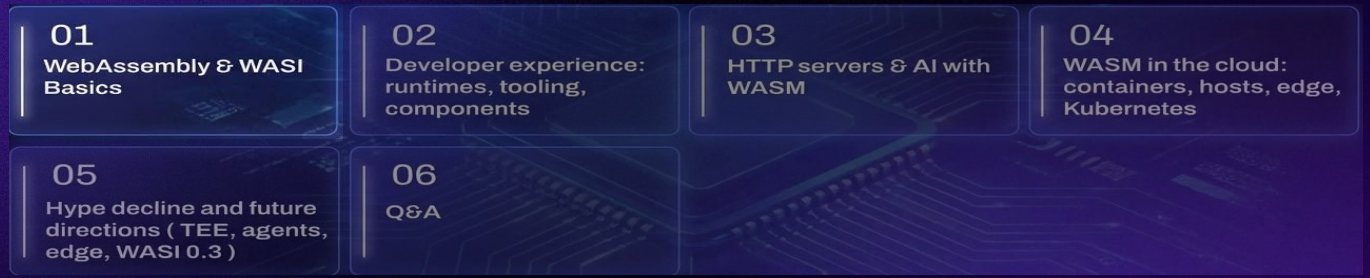
04
WASM in the cloud: containers, hosts, edge, Kubernetes

05
Hype decline and future directions (TEE, agents, edge, WASI 0.3)

06
Q&A

1. WebAssembly Basics

(wasm = WebAssembly)



What is WebAssembly (on the client side) ?

- Designed to fix JavaScript limitations.
- Compact binary format for very efficient execution.
- Performance-intensive Applications

How does it differ from Server Side Wasm ?

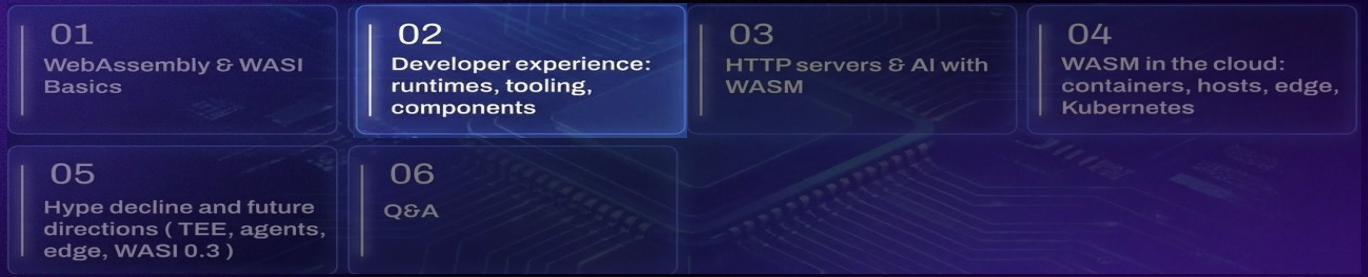
- Compact binary format
- Secure, Sandboxed environment
- Portable
- More efficient than standard linux containers

How does WebAssembly run outside the browser?

- Runtimes allow wasm apps to run
- + WASI (WebAssembly System Interface)

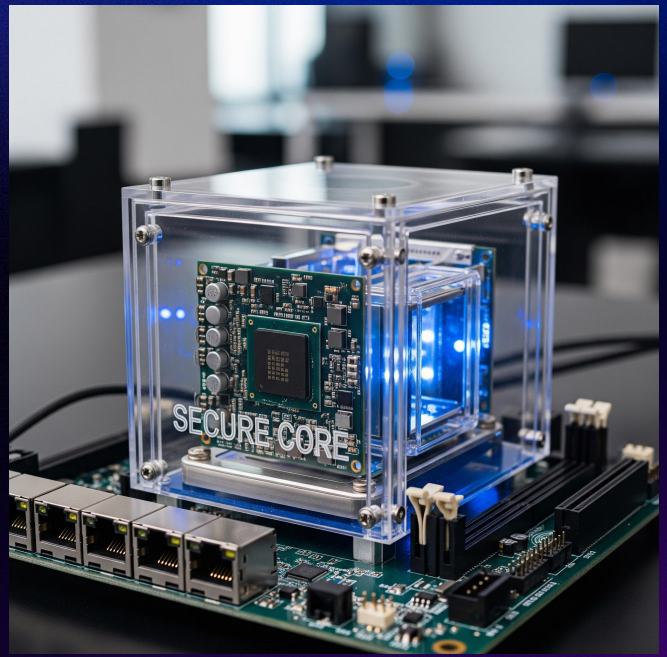
2. Developer experience

(the promises... 🧐)



WASM is Security First… Right?

- Sandboxed, memory-safe modules.
- Explicit imports.
- Security downside = binary difficult to analyze.
- .wat solves this problem



The Cross Language Promise

- Compile once, run everywhere
- Prevent Unauthorized library access
- No need to rebuild the host application.
- Create safe plugins.

Cross Language Interoperability in Practice

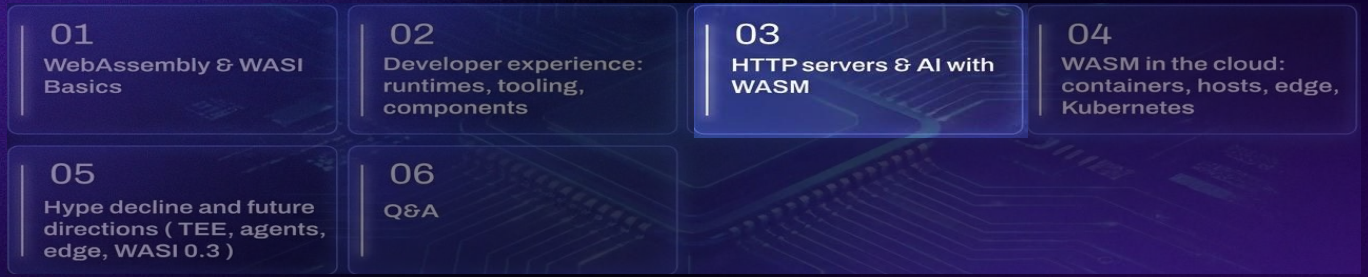
- Rust has great support
- C/C++ has decent support
- Go has mediocre support
- Python has very limited support

Demo one : Running a compiled .wasm file in Java

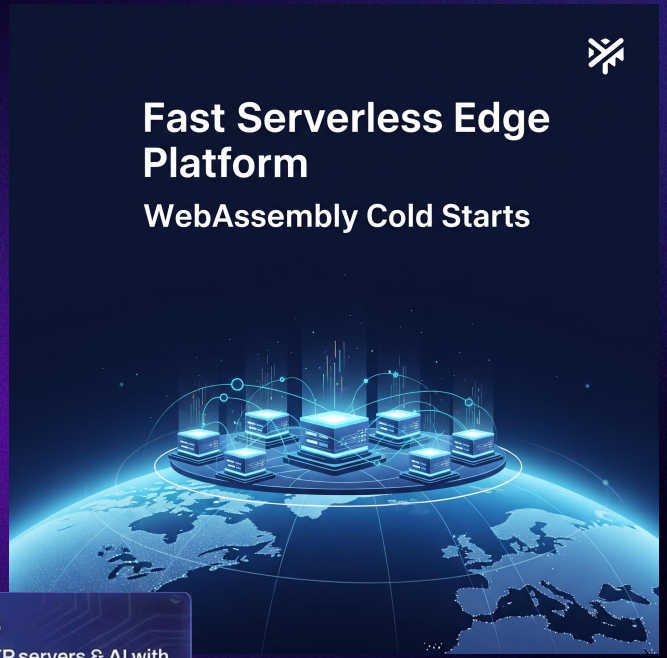
02
Developer experience: runtimes, tooling, components

03 **Next Up**
HTTP servers & AI with WASM

3a. HTTP Servers



Why Wasm for HTTP?



- Portability allows the same file to run on any host.
- HTTP applications are sandboxed with good security.
- Minimal capabilities are required through the WASI interface.
- Extremely fast cold starts benefit serverless and edge platforms.

3b. AI with WASM

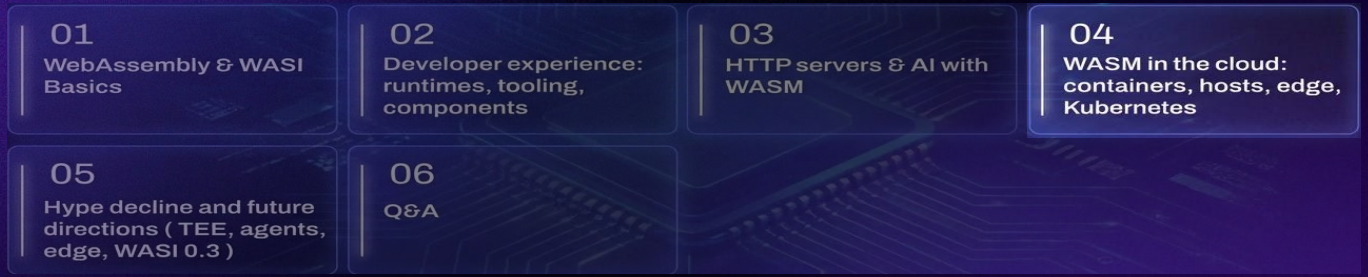
Is WASM helpful in the AI goldrush?

- AI inference logic can be compiled into WebAssembly. (FaaS)
- Execution of Wasm AI models is very fast, often in milliseconds.
- Spin simplify the deployment and execution.

DEMO 2 Moving heavy AI logic + API to WASM.

4. WASM in the sky ☁️

(In the cloud)



Wasm vs. Containers

Wasm

- Lightweight
- Fast Startup
- Language Agnostic
- Secure Sandbox
- Portable
- Small Footprint

Containers

- Isolated Environment
- Operating System Virtualization
- Package Dependencies
- Mature Ecosystem
- Resource-Heavy

Wasm and Kubernetes

- Direct Kubernetes with pure WebAssembly is difficult
- Solutions are either docker-dependent or not maintained
- SpinKube and WasmCloud.

DEMO 3 : WASM Cloud

Wasm at the Edge

01 CDN platforms execute Wasm code close to users.

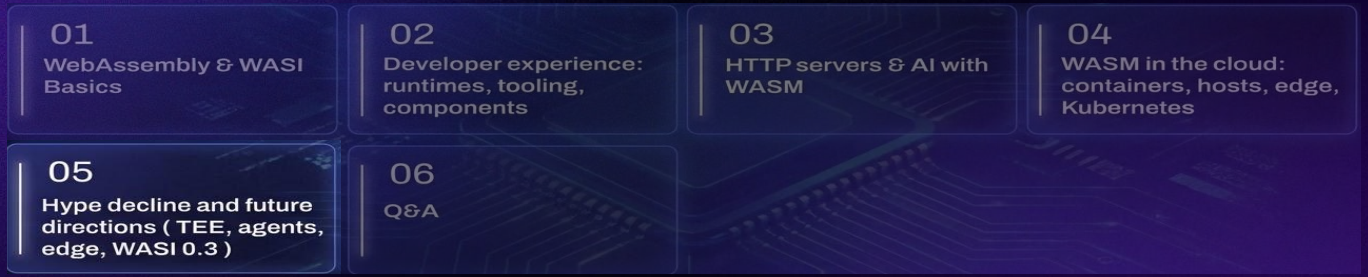
02 Wasm UDFs extend database engines with custom functions.

03 Fast startup times benefit serverless and edge platforms.

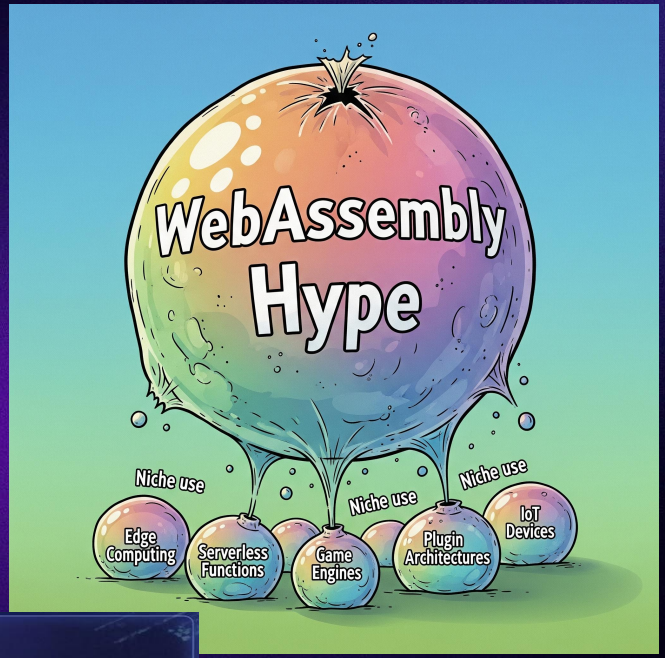
04 Microcontrollers use runtimes like WAMR and wasm3.

04
WASM in the cloud: containers, hosts, edge, Kubernetes

5. Hype decline & future



Wasm Hype vs. Reality



- Early hype suggested Wasm would replace JVM and containers
- Reality shows significant gaps in documentation and tooling
- Market has cooled down significantly for server-side wasm
- wasm's main role is in niches and with big companies

Wasm's Current Wins

01 Cross-language interoperability creates safe plugin architectures.

02 It is a key solution for confidential computing and AI agents.

03 Fast cold starts benefit serverless and edge workloads.

04 Wasm is useful for rewriting microservices as FaaS.

05
Hype decline and future directions (TEE, agents, edge, WASI 0.3)

The Future of wasm?

01 Wasm and TEEs enable secure confidential computing.

02 WASI 0.3 features will enable concurrent I/O and streams.

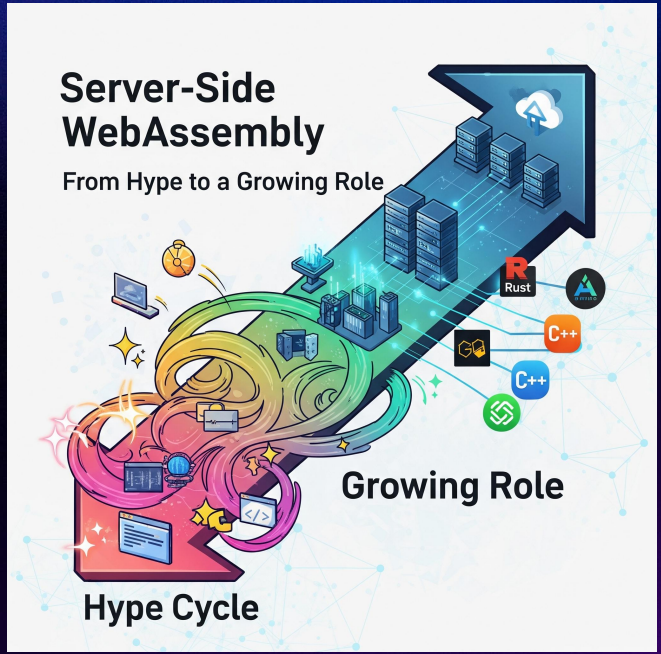
03 AI agents use sandboxing to limit prompt injection risks.

04 Edge WAF logic enhances security with reduced latency.

05 Hype decline and future directions (TEE, agents, edge, WASI 0.3)

Conclusion Summary and Future

- From hype to a growing role.
- Strong fit for edge/FaaS, plugins, and AI agents.
- Documentation, adoption, and implementation still severely lack.



The End

- Questions about our presentation are welcome
- Visit the website!

06
Q&A